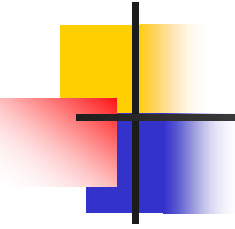


11-3 HDLC

High-level Data Link Control (HDLC) is a bit - oriented protocol for communication over point-to-point and multipoint links. It implements the Stop-and-Wait protocol we discussed earlier. Although this protocol is more a theoretical issue than practical, most of the concept defined in this protocol is the basis for other practical protocols such as PPP, Ethernet, or wireless LANs.



111.3.1 Transfer Modes

HDLC provides two common transfer modes that can be used in different configurations: normal response mode (NRM) and asynchronous balanced mode (ABM).

Figure 11.14: *Normal response mode*

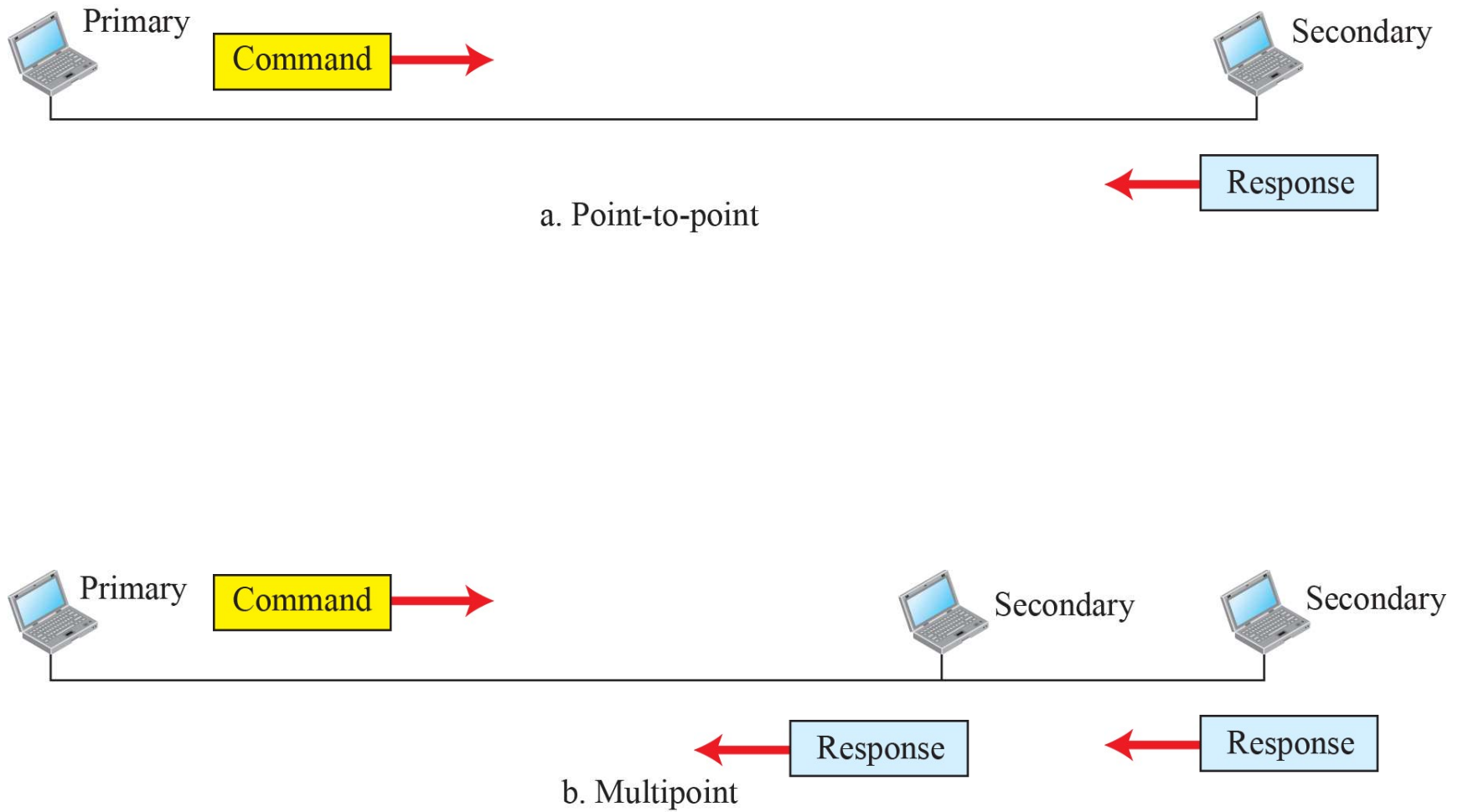
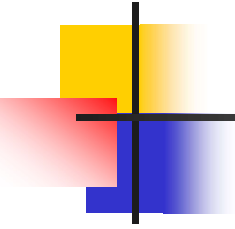


Figure 11.15: Asynchronous balanced mode





11.3.2 Framing

To provide the flexibility necessary to support all the options possible in the modes and configurations just described, HDLC defines three types of frames: information frames (I-frames), supervisory frames (S-frames), and unnumbered frames (U-frames).

Figure 11.16: HDLC frames

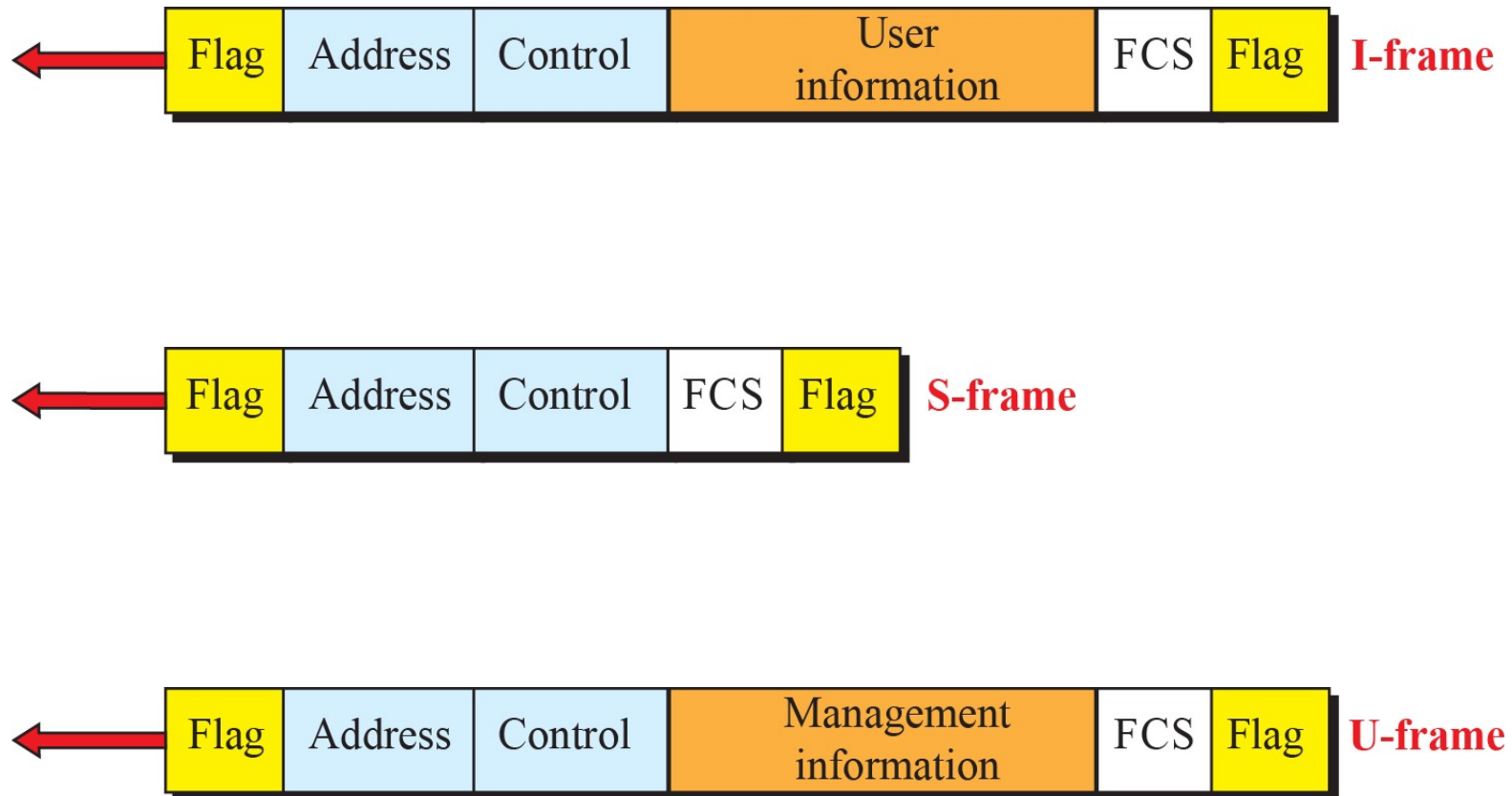
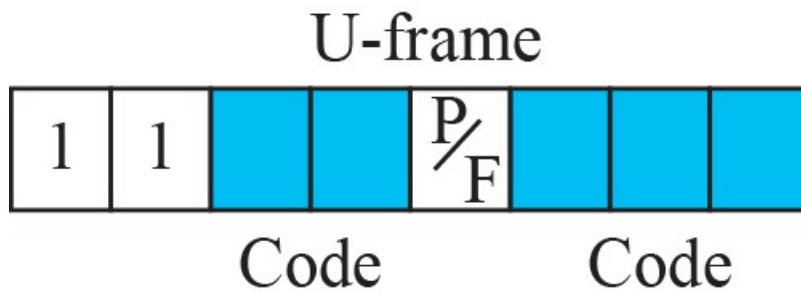
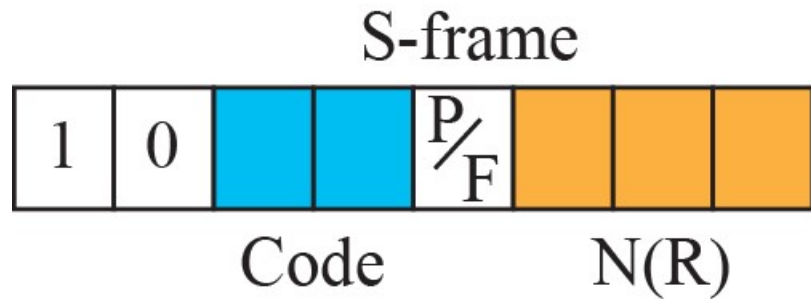
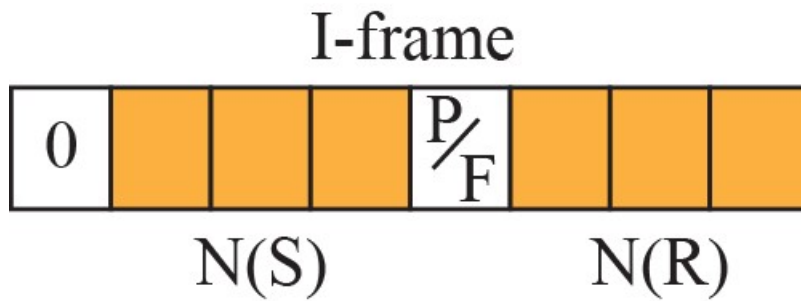


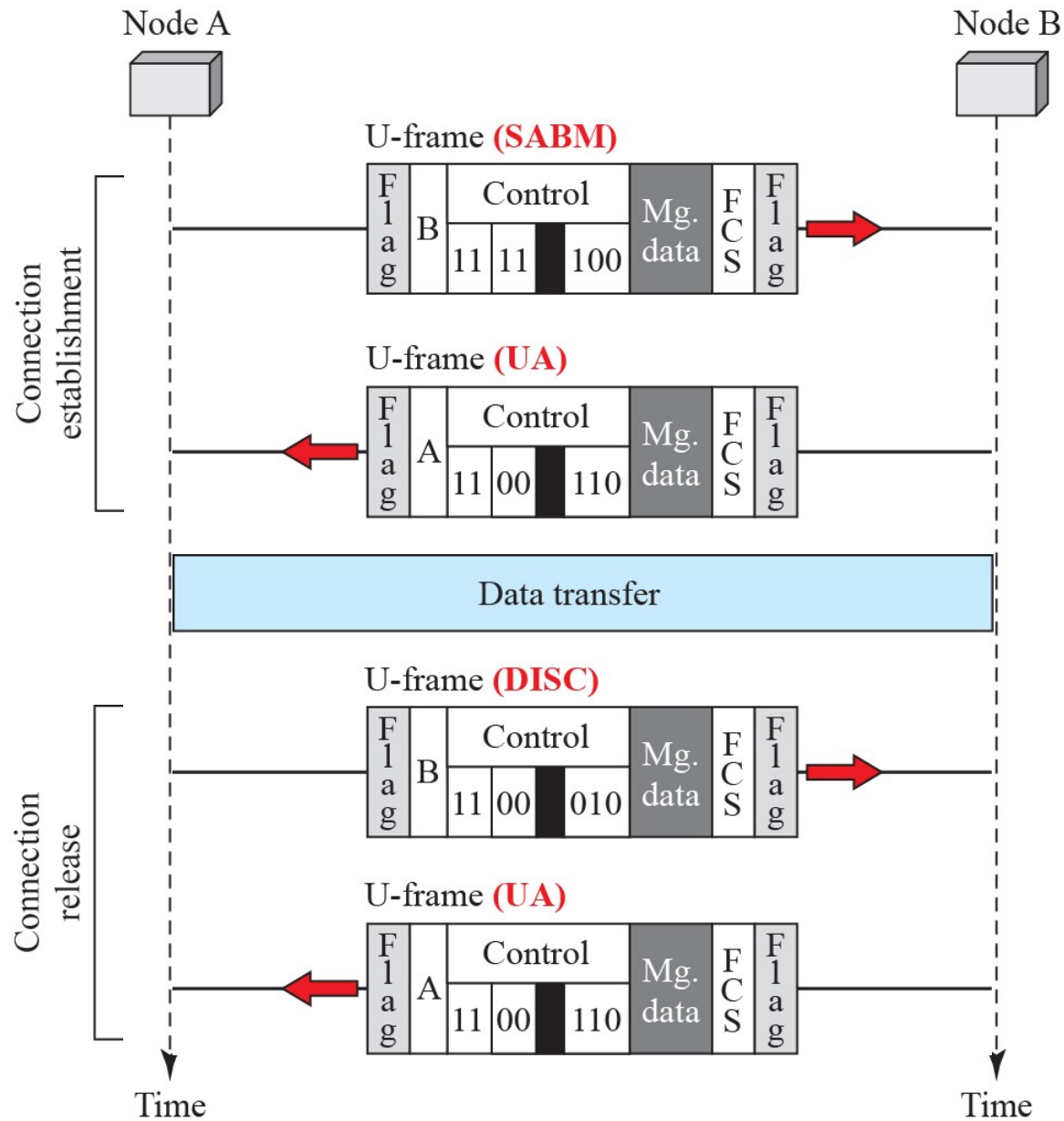
Figure 11.17: Control field format for the different frame types



Example 11.5

Figure 11.18 shows how U-frames can be used for connection establishment and connection release. Node A asks for a connection with a set asynchronous balanced mode (SABM) frame; node B gives a positive response with an unnumbered acknowledgment (UA) frame. After these two exchanges, data can be transferred between the two nodes (not shown in the figure). After data transfer, node A sends a DISC (disconnect) frame to release the connection; it is confirmed by node B responding with a UA (unnumbered acknowledgment).

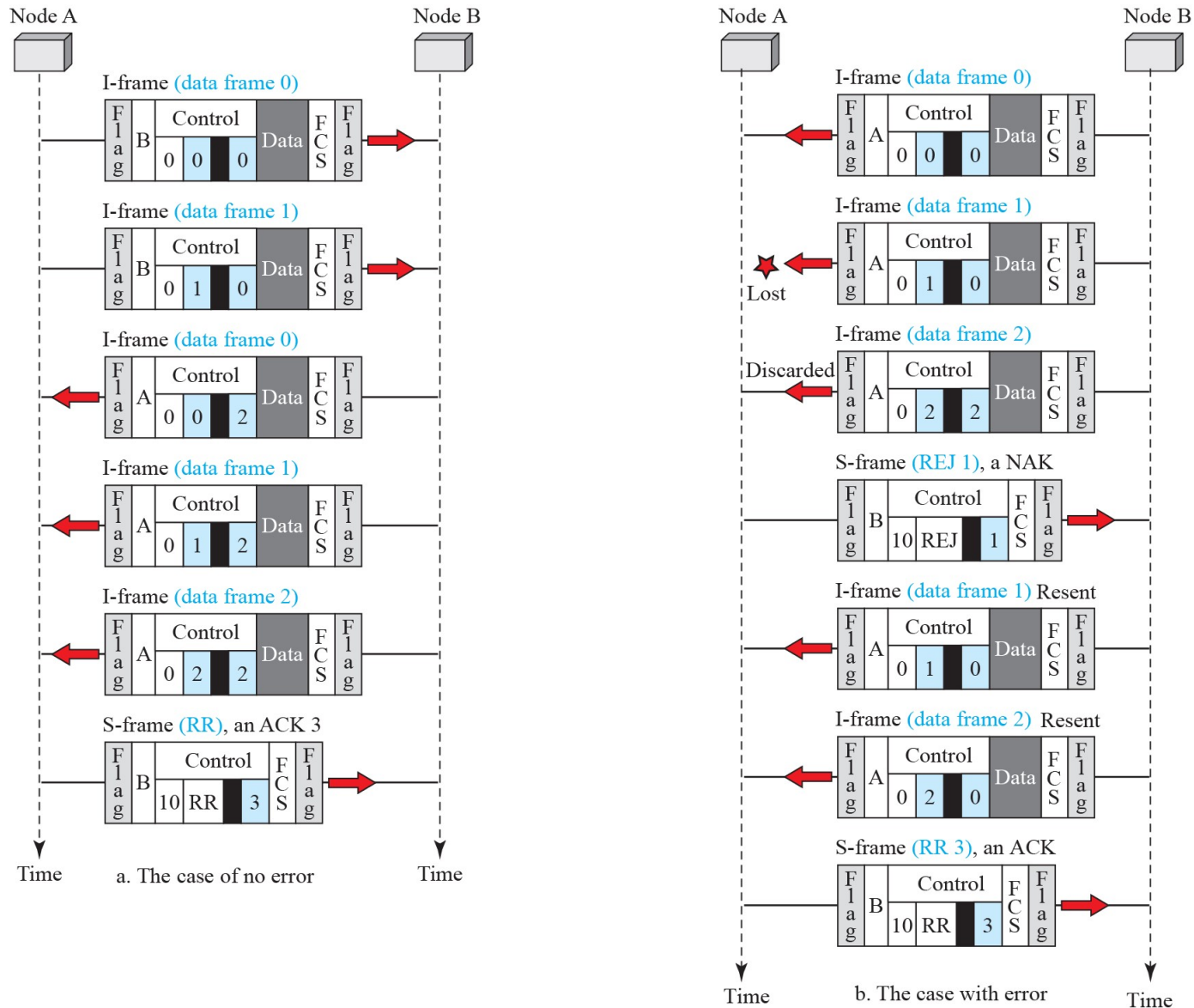
Figure 11.18: *Example of connection and disconnection*



Example 11.6

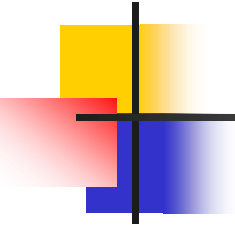
Figure 11.19 shows two exchanges using piggybacking. The first is the case where no error has occurred; the second is the case where an error has occurred and some frames are discarded.

Figure 11.19: Example of piggybacking with and without error



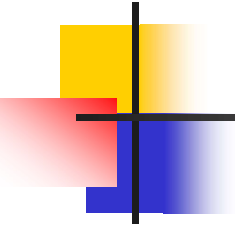
11-4 PPP

One of the most common protocols for point-to-point access is the Point-to-Point Protocol (PPP). Today, millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP. To control and manage the transfer of data, there is a need for a point-to-point protocol at the data-link layer. PPP is by far the most common.



11.4.1 Services

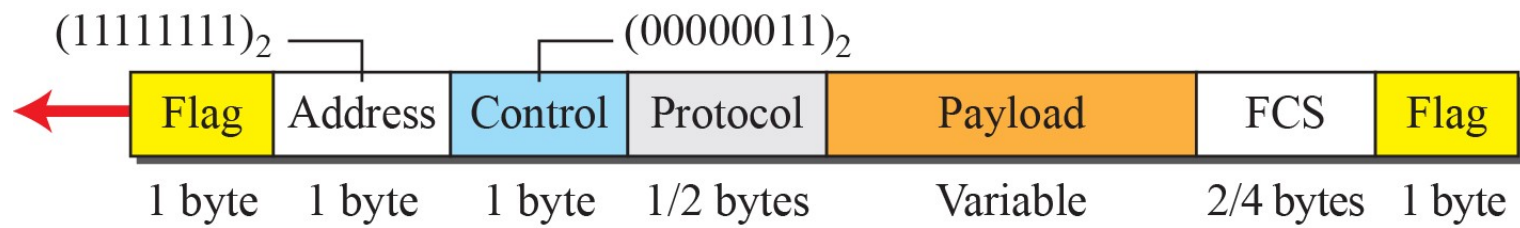
The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple.

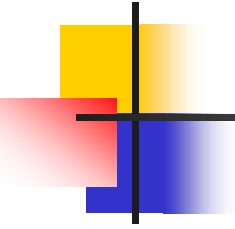


111.4.2 Framing

PPP uses a character-oriented (or byte-oriented) frame. Figure 111.20 shows the format of a PPP frame.

Figure 11.20: PPP frame format

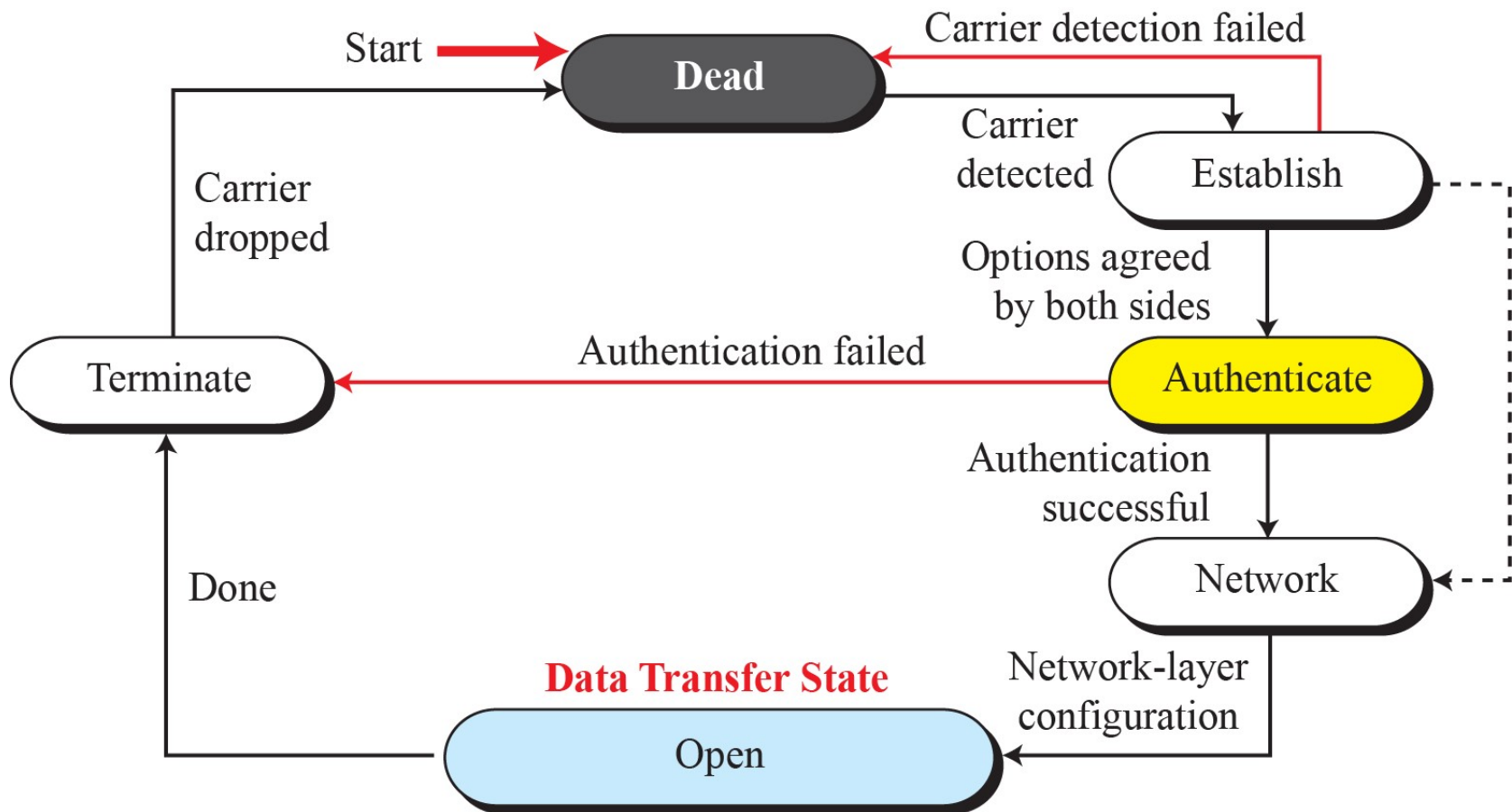


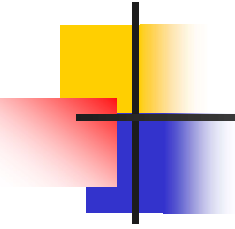


111.4.3 Transition Phases

A PPP connection goes through phases which can be shown in a transition phase diagram (see Figure 111.21).

Figure 11.21: Transition phases

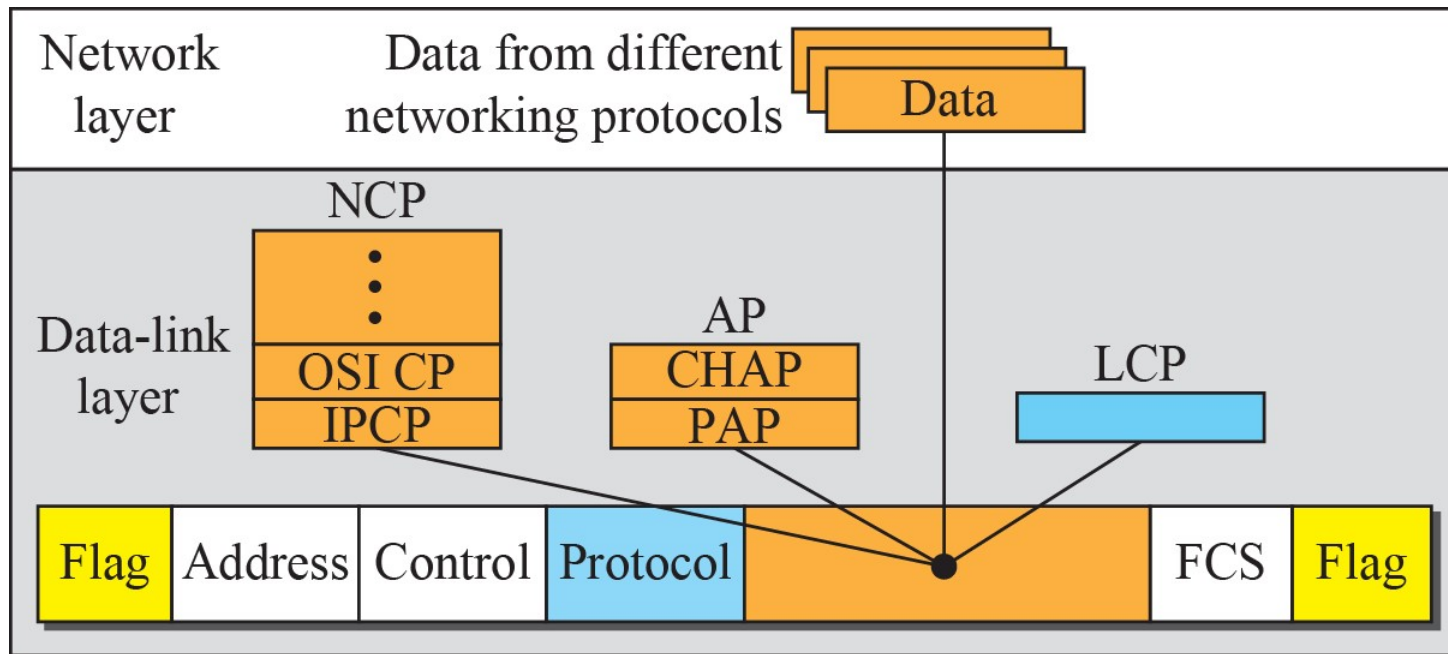




11.4.4 Multiplexing

Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate the parties involved, and carry the network-layer data. Three sets of protocols are defined to make PPP powerful: the Link Control Protocol (LCP), two Authentication Protocols (APs), and several Network Control Protocols (NCPs).

Figure 11.22: Multiplexing in PPP



Legend

LCP : Link control protocol
AP : Authentication protocol
NCP: Network control protocol

Protocol values:

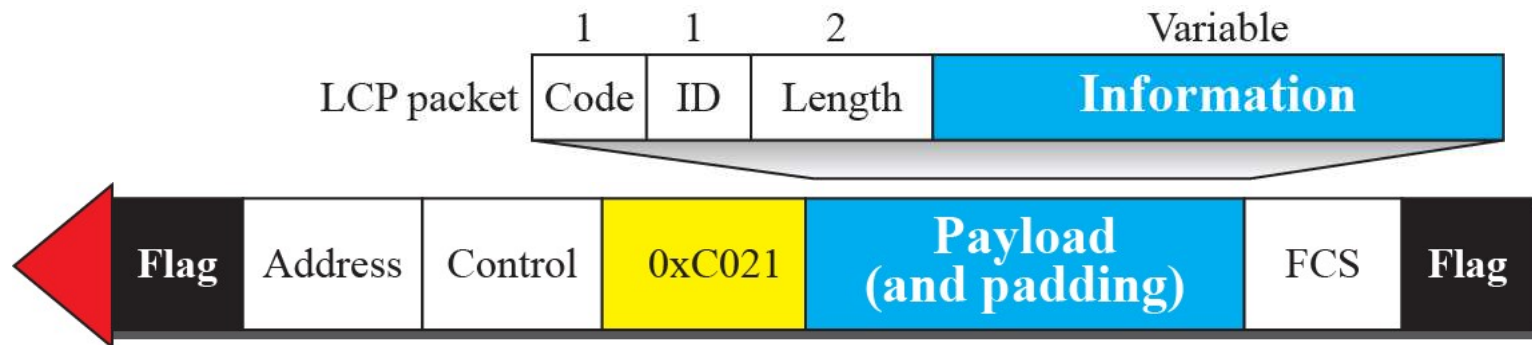
LCP : 0xC021

AP : 0xC023 and 0xC223

NCP: 0x8021 and

Data: 0x0021 and

Figure 11.23: LCP packet encapsulated in a frame



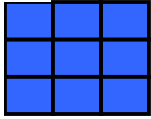


Table 11.1: LCP Packets

Code	Packet Type	Description
0x01	Configure-request	Contains the list of proposed options and their values
0x02	Configure-ack	Accepts all options proposed
0x03	Configure-nak	Announces that some options are not acceptable
0x04	Configure-reject	Announces that some options are not recognized
0x05	Terminate-request	Request to shut down the line
0x06	Terminate-ack	Accept the shutdown request
0x07	Code-reject	Announces an unknown code
0x08	Protocol-reject	Announces an unknown protocol
0x09	Echo-request	A type of hello message to check if the other end is alive
0x0A	Echo-reply	The response to the echo-request message
0x0B	Discard-request	A request to discard the packet

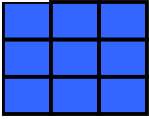


Table 11.2: Common options

<i>Option</i>	<i>Default</i>
Maximum receive unit (payload field size)	1500
Authentication protocol	None
Protocol field compression	Off
Address and control field compression	Off

Figure 11.24: PAP packets encapsulated in a PPP frame

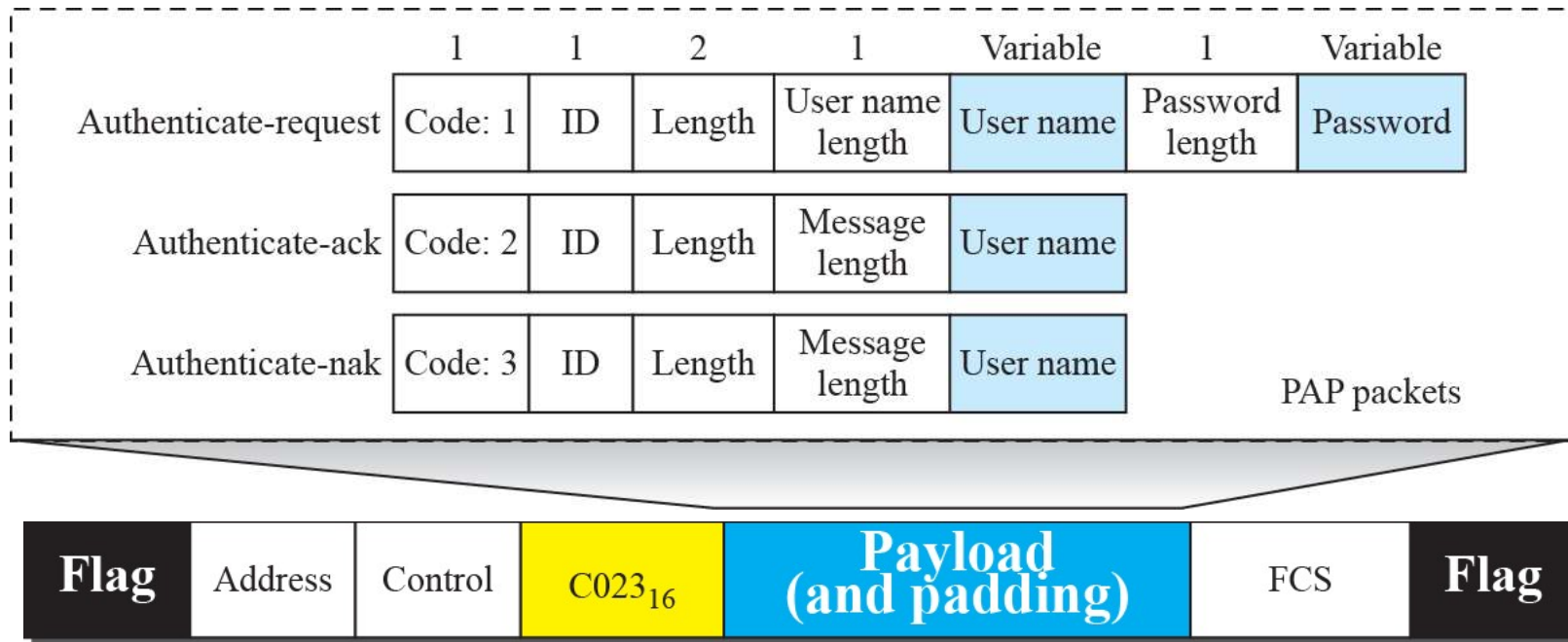
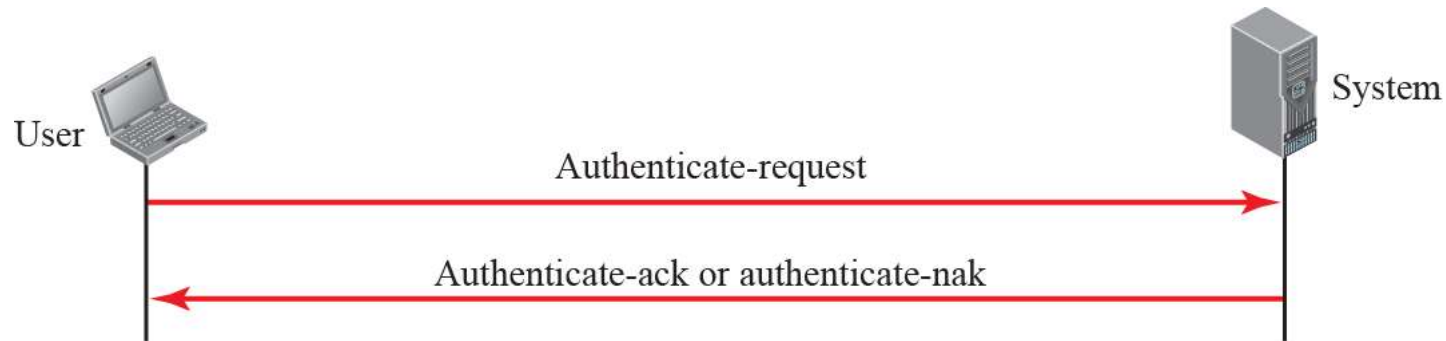


Figure 11.25: CHAP packets encapsulated in a PPP frame

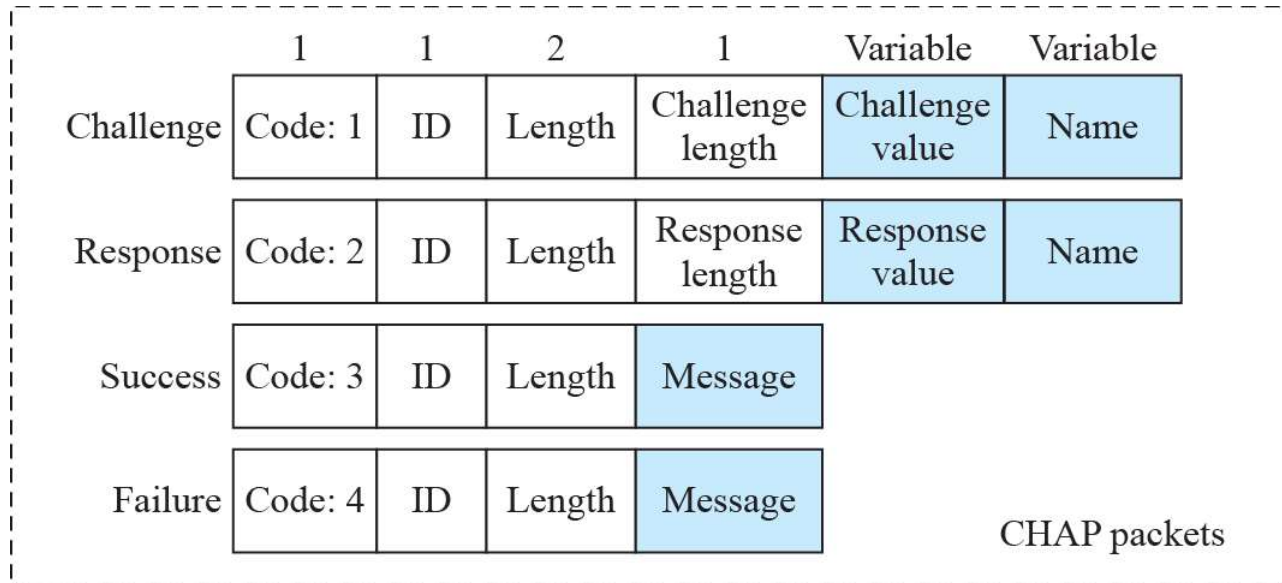
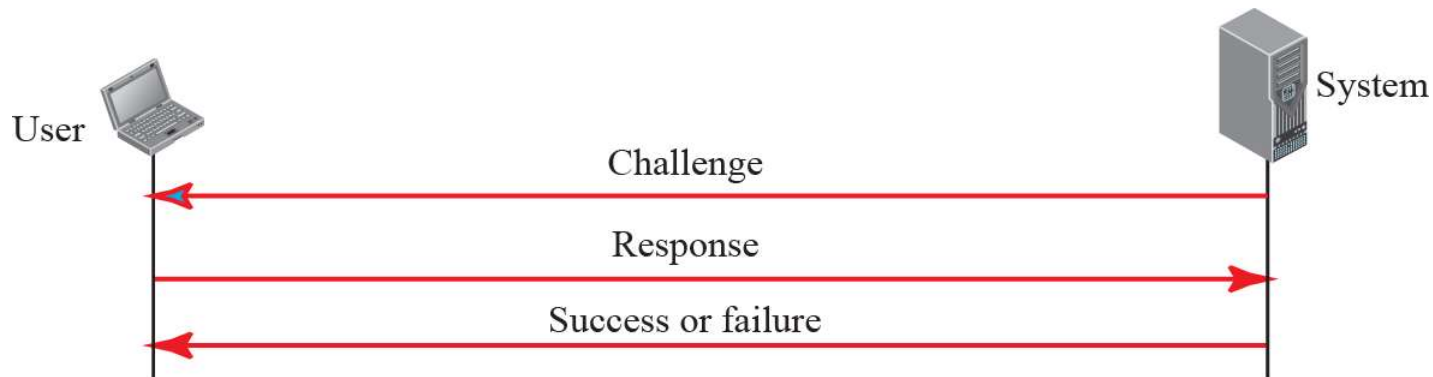
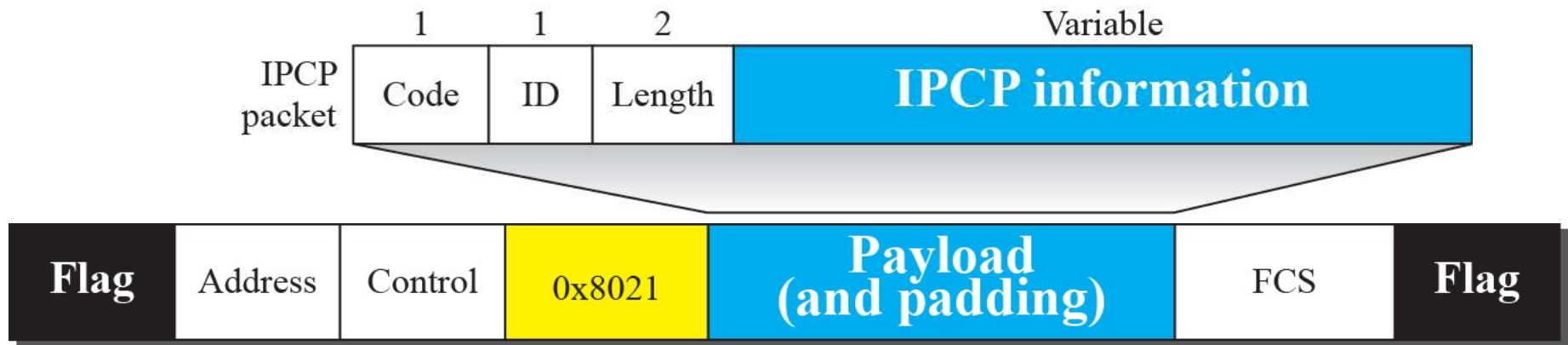


Figure 11.26: *IPCP packet encapsulated in PPP frame*



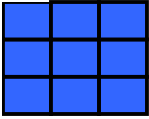


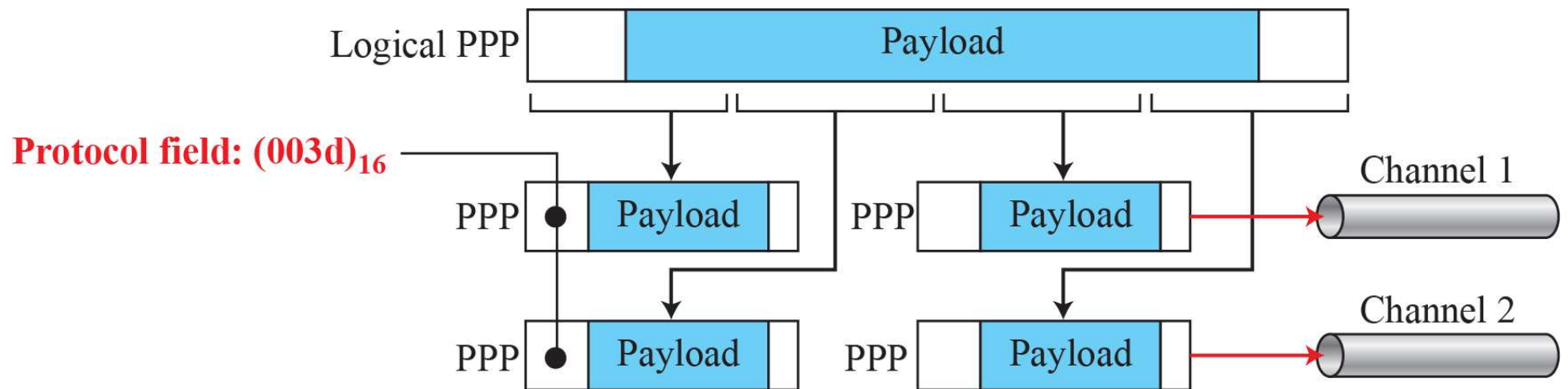
Table 11.3: Code values for IPCP Packets

<i>Code</i>	<i>IPCP Packet</i>
0x01	Configure-request
0x02	Configure-ack
0x03	Configure-nak
0x04	Configure-reject
0x05	Terminate-request
0x06	Terminate-ack
0x07	Code-reject

Figure 11.27: IP datagram encapsulated in a PPP frame



Figure 11.28: Multilink PPP



Example 111.7

Let us go through the phases followed by a network layer packet as it is transmitted through a PPP connection. Figure 111.29 shows the steps. For simplicity, we assume unidirectional movement of data from the user site to the system site (such as sending an e-mail through an ISP).

Figure 11.29: An example

